# The Analysis Stage

## From a research question to a result you can defend

With an example on hydropower dams and conflict

D'Amico (2026), *Global Environmental Politics*

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# Today

1. What analysis is, and the main ways of doing it
2. From correlation to causation
3. Analysis: dams and conflict
4. Beyond staggered DiD: does your thesis have a "treatment"?
5. Questions, then a lab in R, in your browser

**The one idea to take away**

A result is the end of a chain of choices. Good analysis shows those choices, and checks whether the result changes if you make them differently.

# Where we are in the course

**1 Writing a thesis**

Week 1

Break it into pieces and set deadlines you commit to

**2 Developing an idea**

Week 2

What interests you? What questions can actually be asked?

**3 Literature review**

Week 3

Find the puzzle: what has not been done yet, and what you add

**4 Concepts and theory**

Weeks 4-5

The "why": why you expect what you expect

**5 Research design**

Weeks 6–7

A design that matches the question

**6 Conducting the research**

Week 9, and today's lab

**The hard part: gathering and running it**

**7 Analysis**

Today

**Interpretation, discussion, and sometimes more analyses**

**8 Writing the paper**

Week 11

Full circle: was the question answered? Limits, surprises, and what comes next

# 01
# What is analysis?

# More than listing

> "Analysis is more than listing, more than description, more than assertion, more than a compilation of everything a researcher has done."
>
> Franklin, Chapter 7

**Commit**
Take a stance before you start (deduction), build one as you go (induction), or mix the two

**Think**
Decide what the material means and what counts as evidence

**Do**
Process, represent, and present it, and say what you left out

Franklin warns against **"research design by prejudice rather than consideration"**: picking quantitative or qualitative first, then forcing the question to fit.

# Quantitative and qualitative analysis

Franklin puts the divide between seeing "the whole as a sum of its parts" and seeing it as "greater than the sum of its parts".

| | Quantitative | Qualitative |
|---|---|---|
| Typical question | How much? How often? Does X change Y? | How? Why? What does it mean? |
| Material | Counts, surveys, events, many cases | Texts, interviews, images, few cases |
| Reasoning | Mostly deductive | Often inductive |
| Content | **Manifest**: what is said. Count it (how often "security" appears) | **Latent**: what is meant. Interpret it (how a group is portrayed) |
| Strength | Patterns and effects across many units | Mechanisms and context |

In practice the line is blurry. A point from your comments: counting words or events can be the *first step* to interpretation, and a good quantitative paper still has to interpret.

# Common ways to analyse qualitative material

**Content analysis**
Code texts into categories, then count or compare them. Works on *manifest* content: what is said

**Thematic coding**
Build themes from interviews or documents, often inductively (grounded approaches)

**Discourse analysis**
How language frames a problem, and who gets to define it

**Textual and visual analysis**
Close reading for *latent* meaning: what is meant, in context

**Process tracing**
Follow the steps from cause to outcome in one case. A strong way to get at the "why"

**Keeping it honest**
A written codebook, a second coder on a sample, and reporting where coders disagree

Content, textual, and discourse analysis are the three Franklin covers in Chapter 7.

# Why I do quantitative work

- My questions are about **many places over many years**: where does conflict rise or fall after a shock?
- Data now cover the whole world at a fine scale: conflict events, dams, disasters, food prices
- Quantitative designs let me ask what would have happened **without** the shock

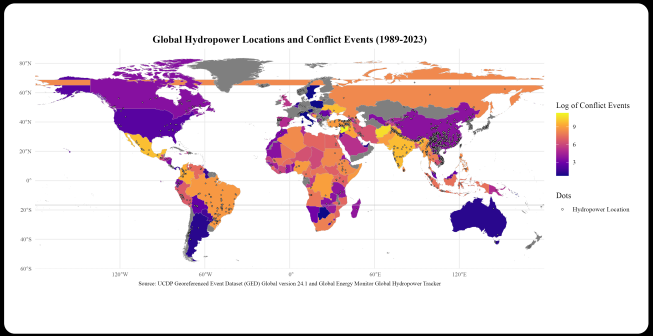
Even with statistics, Franklin says, "the researcher still needs to engage with these results by making an **interpretative intervention**."

Numbers do not speak for themselves. Choosing the unit, the comparison, and the outcome is already interpretation.

# 02

# From correlation to causation

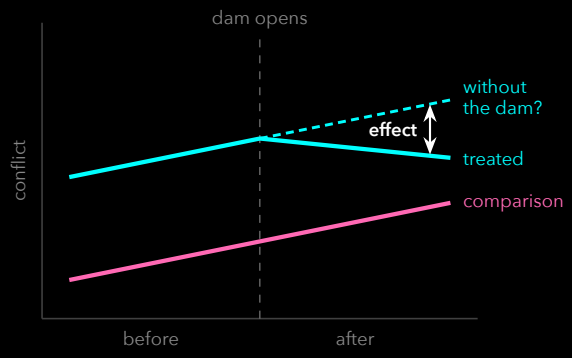
# Dams and conflict on a map




- Dams and conflict often sit in the same places
- But dams are **not built at random**: rivers, terrain, money, and state reach all matter
- Those may also affect conflict
- Conflict may also stop dams being built

So the overlap on the map is a correlation. It is not yet an effect.

# The question behind every causal claim




- We never see what a place would have looked like **without** its dam
- Difference-in-differences borrows that path from places that did not get a dam (yet)
- Effect = change in treated places **minus** change in comparison places

**Parallel trends:** without the dam, both groups would have moved together. We check it by looking at the years *before*.

# Why this kind of DiD: dams arrive in different years

- The textbook version has one treatment date. Dams open in 1990, 2001, 2015…
- The old one-regression approach (two-way fixed effects) quietly uses **already-treated** places as the comparison
- Callaway and Sant'Anna (2021) compare each group only with places **not yet treated**

| Country | Old way | Callaway–Sant'Anna |
|---|---|---|
| Pakistan | $+25.69$ | $\mathbf{-19.59}$* |
| Sudan | $+13.18$ | $+7.32$* |
| Colombia | $-2.72$ | $-4.81$* |
| Philippines | $-1.13$* | $-1.01$* |

Effect on conflict events per grid cell-year. * 95% interval excludes zero. D'Amico (2026), Table A.I.

> In Pakistan the sign flips. The comparison you choose can change the answer.

# 03

# Analysis: dams and conflict

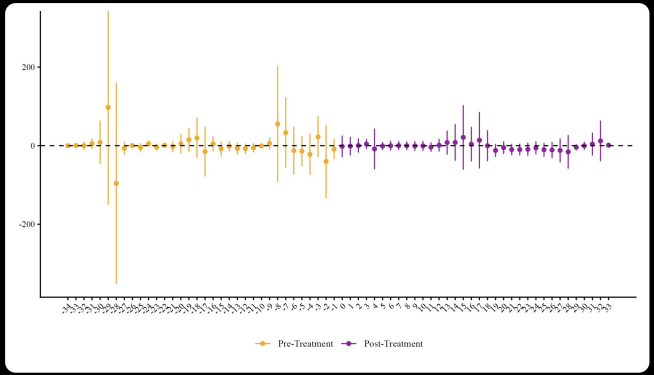
# The design, one choice at a time

| Choice | What the paper does | Could have been… |
|---|---|---|
| Unit | Hexagonal grid cells, 1.25° (about 140 km across) | Districts, provinces, finer cells |
| Treatment | First dam of 75 MW or more (Global Hydropower Tracker) | Any dam, capacity, dam status |
| Outcome | Armed conflict events (UCDP) | Deaths, civilian deaths |
| Comparison | Cells not yet treated | Cells never treated |
| Timing | Effects may start a year early (construction) | No anticipation |
| Cases | 8 countries, 1989–2024 | Pooled across all countries |

Franklin calls this "making a commitment". Every row is one.

**Why hexagons?** (from your comments) Same shape and size everywhere, and no borders drawn for political reasons. 1.25° is "a balance between granularity and coherence": smaller cells add zeros and noise, bigger ones blur places together.

# Pooled across all eight countries: nothing

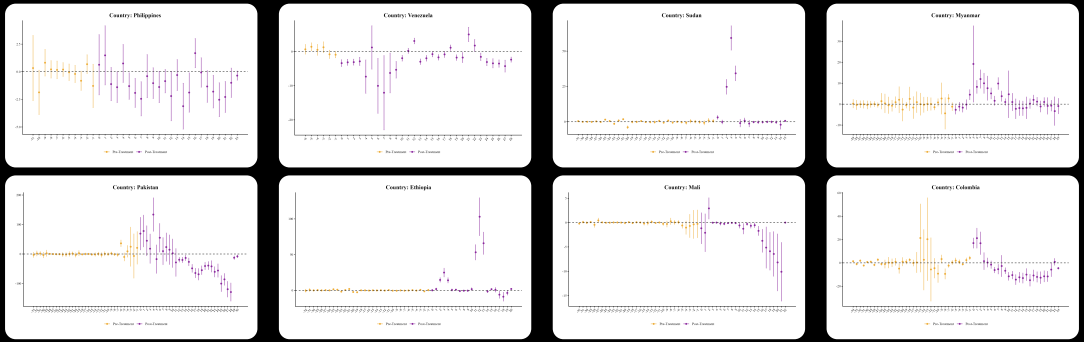



- **x** years before and after the first dam
- **y** effect on conflict events
- **Orange** before, **purple** after
- Estimates bounce around zero

> Stop here and the conclusion is "dams have no effect on conflict".

D'Amico (2026), *Global Environmental Politics*, Figure 2.

# Country by country: three different stories




**Falls** Philippines, Venezuela **Rises, then returns** Sudan, Myanmar **Disruption, then falls** Pakistan, Ethiopia, Mali, Colombia

# So what can a policymaker learn?

From your comments: if the pooled result is zero and countries differ, what is the general lesson?

- The pooled zero hides real effects that cancel out
- "Dams cause conflict" and "dams bring peace" are both too simple
- The useful question becomes: **where** and **under what conditions**?

DiD tells you *whether* and *when* conflict changed. It cannot tell you *why*.

Governance, who gets displaced, who gets the electricity and the jobs: that is where case studies and process tracing come in.

The data do not speak for themselves (Franklin). Back to quantitative *and* qualitative.

# Robustness: does the result hold up?

| Country | Choice changed | Before | Alternative | After | Verdict |
|---|---|---|---|---|---|
| Colombia | Comparison group | $-4.81$* | never treated | $-4.80$* | Robust |
| Venezuela | Comparison group | $-2.51$* | never treated | $-2.51$* | Robust |
| Colombia | Anticipation | $-4.81$* | none | $-7.65$* | Same sign, bigger |
| Venezuela | Anticipation | $-2.51$* | none | $-1.60$* | Same sign, smaller |
| Philippines (disasters) | Anticipation | $+2.37$* | one year | $+1.05$ | Fragile |

Effect on conflict events per grid cell-year. * 95% interval excludes zero. My reruns of the paper's model.

**Robust:** same sign, similar size. **Fragile:** the sign flips, the size jumps, or the interval now includes zero.

Change **one** thing at a time, and only to choices you could defend. You do this in Step 4 of the lab.

# What I'd look at more carefully now

| Country | With a dam | Used |
|---|---|---|
| Philippines | 3 | **1** |
| Venezuela | 4 | **1** |
| Colombia | 14 | 7 |
| Mali | 2 | 2 |
| Sudan | 3 | 3 |
| Ethiopia | 8 | 7 |
| Pakistan | 11 | 10 |
| Myanmar | 22 | 21 |

- Dams that opened before 1991 have no "before" period, so the method drops those cells
- The Philippines and Venezuela results each rest on **one** grid cell
- Some dams are only planned or under construction. All start 2022–24, so they barely matter here, but I would flag it

> From your comments: it is easy to treat a significant coefficient as the conclusion. This is why: the asterisk says nothing about how many places stand behind it.

# Same design, different outcome

| Country | Conflict events | Total deaths |
|---|---|---|
| Sudan | $+7.32$* | $+76.14$* |
| Myanmar | $+2.41$* | $+26.20$ |
| Pakistan | $-19.59$* | $-248.68$* |
| Ethiopia | $+12.57$* | $-41.64$ |

Effect per grid cell-year, paper's specification. * 95% interval excludes zero.

- "Conflict" can mean how often fighting happens or how deadly it is
- In Ethiopia the sign flips, with a huge margin of error: a few war years drive the deaths
- Neither outcome is *the* right one. Which you report, and why, is part of the analysis

# 04
# Beyond staggered DiD

# When everyone is treated at once

A thesis example. When the US left the Paris Agreement, did the small-island coalition speak up more in UN climate talks?

| % of all speaking turns | US in | US out | Change |
|---|---|---|---|
| United States | 5.17 | 2.12 | −3.05 |
| Small islands | 4.09 | 4.71 | +0.62 |
| EU (comparison) | 7.09 | 7.54 | +0.45 |
| **DiD:** islands' change minus EU's | | | ≈ 0 |

| % of loss-and-damage turns | US in | US out | Change |
|---|---|---|---|
| United States | 4.42 | 0.00 | −4.42 |
| Small islands | 3.88 | 17.60 | +13.72 |
| EU (comparison) | 0.88 | 4.65 | +3.77 |
| **DiD:** islands' change minus EU's | | | ≈ +10 |

19 negotiating meetings after Paris (11 US in, 8 during withdrawal). DiD estimates $+0.18$ (SE 0.92) and $+9.95$ (SE 10.77). A quick illustration, not the thesis model.

Everyone is hit at once, so no group is "not yet treated". The comparison is a group that is **less exposed**: a classic 2×2 DiD.

- **First, check the treatment happened.** The US spoke far less
- **Overall:** the islands did not speak up more than the EU did
- **On loss and damage:** a big jump, but the uncertainty is as large as the effect. Promising, not proven

# Which kind of DiD fits your question?

| Your treatment… | Example | Design |
|---|---|---|
| hits different units at **different times** | Dams, minimum wage rises | Staggered DiD (Callaway–Sant'Anna) |
| hits **everyone at once** | US leaves Paris, Brexit vote | 2×2 DiD on more vs less exposed units |
| switches **on and off** | A policy reversed, then restored | Possible, but harder |
| does not exist | "Is civil disobedience justified?" | Not DiD, and that is fine |

Start from the question, then pick the design. Doing it the other way round is Franklin's "research design by prejudice". Step 6 of the lab asks you this about your own thesis.

# Discussion: could disasters be the treatment?

Same grid, same method, different treatment. Instead of a cell's first **dam**, it is the cell's first **flood, storm, or other climate disaster** (EM-DAT).

- No one *chooses* where a flood hits. Is that better than dams?
- EM-DAT gives map coordinates for only about one event in four, and only for 2001–2020
- Philippines: $+2.37$* with no anticipation, $+1.05$ (not distinguishable from zero) with one year

**Talk to your neighbour**

1. Some "never treated" cells were probably hit too. What does that do to the comparison?
2. Floods hit the same places again and again. Is the first one the right treatment?
3. How would you measure the treatment better?

You can try it yourself in Challenge E.

# Questions

## What would you like to ask?

From your comments on the readings:

- How confidently can qualitative work explain the "why"?
- How do you keep bias out of a coding scheme?
- Is Franklin too harsh on counting?
- How do you pick the unit of analysis?

# 05
# Lab

# The lab, step by step

| | Step | You do | Look for |
|---|---|---|---|
| 0 | Look at the data | Open the panel and the dam list | Cells that appear twice |
| 1 | Build the treatment | With dplyr: each cell's first dam, merged onto the panel | Countries with very few treated cells. Which dams count? |
| 2 | Reproduce the paper | Run the paper's model for the Philippines | $-1.007$? Flat before year 0? |
| 3 | Explore | `explore()` 3+ countries, outcomes, and disasters | Usable cells, spread of dam years, parallel trends |
| 4 | Your analysis | Pick a country, then change one choice at a time | Robust or fragile? |
| 5 | Challenge | Pick one of A–E. No code given | One honest sentence for a paper |
| 6 | Your thesis | On your own: is there a treatment? Which DiD fits? | What would change your mind |

# The lab: R in your browser, nothing to install



**www.elisadamico.net/analysis-stage-lab**

Code and data: github.com/elisadamico/analysis-stage-lab

- Run it **in order from the top**. R only knows what you have already run
- Steps 0–5 **in pairs**: one person who has coded before with one who has not. **The one who has not types**
- Step 6 **on your own**
- Run a line with `Ctrl`/`Cmd` + `Enter`. The guide at the top of the page explains every button
- Write answers after `# ANSWER:`
- **Download your script** at the end
- Then click Give feedback. It is anonymous